Greecon
SH.P.K

14

HARDWARE SHEET HW14 · QUALIFIED BY ITS REGISTER TABLE

Temperature and humidity sensor

The sensor that reads the air at the field. The platform calls it the Weather sensor; its frost alert uses the grower's own number.

QUALIFIES	Qualified by its register table. An air temperature and humidity sensor that speaks Modbus RTU on RS-485, whose maker publishes its register table, and whose number and speed can be set.
USED IN	Kit F (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager at a farm or a greenhouse
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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Greecon Sh.p.k · Durana Tech Park · info@greecon.earth

HW14

Temperature and humidity sensor

The temperature and humidity sensor reads the air at the field or in the greenhouse. The platform calls it the **Weather sensor**. It speaks Modbus RTU on the same data cable as everything else. **No model is approved yet,** so it qualifies by its maker's register table: its address is entered by hand, and its two readings are mapped by hand. Its most useful work is the frost alert, and that uses the grower's own number.

At a glance

WHAT IT DOES	Measures air temperature and humidity at the field. Unit 7, mapped by hand from the maker's register table.
HOW IT QUALIFIES	**Qualified by its register table.** An air temperature and humidity sensor that speaks Modbus RTU on RS-485, whose maker publishes its register table, and whose number and speed can be set.
SPECIFICATION	RS-485 Modbus, with a published register table
NUMBER ON THE CABLE	7. Set the maker's way, at the office, before the sensor is mounted.
FITTED BY	The project manager.
PLANNING ESTIMATE (EUR)	20 to 80. Not a quote.
USED IN	Kit F. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ It speaks **Modbus RTU on RS-485,** on two data wires.
- ☐ **Its maker publishes its register table,** and you have it before you buy: the register of the temperature and of the humidity, the kind of number each is, and its scale.
- ☐ **The table says how a temperature below zero is given.** It is usually a signed 16-bit number.
- ☐ Its unit number and its speed can be set, and the maker says how: to a number of its own, and to 9600 baud, 8 data bits, no parity, 1 stop bit.
- ☐ It is made for where it will hang: outdoors, a sensor with a shield against sun and rain.
- ☐ Its supply voltage is one the kit has: read its label against the 24 V supply.

STOP **Do not buy a sensor whose register table you have not seen.** A sensor whose maker does not document its registers cannot be mapped honestly, so it cannot be used.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	None.
S Solar plant with grid connection	NOT USED	None	None.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, at the field or in the greenhouse. It gives Air temperature and Humidity. The frost alert says nothing until the grower's own Frost alert at or below value is entered.
W Water tank and pump station	NOT USED	None	None.

What Greecon reads from it

Reading in Greecon	Unit	What is real	Standard alert
Air temperature	°C	-40 to 70	None. *Frost alert at or below* has **no standard value**
Humidity	%	0 to 100	None

The frost alert. Frost belongs to the crop: the same air is harmless to one and ruinous to another, and it depends on how far on the crop is. So nobody can set the limit for the grower, and until it is entered Greecon says nothing about frost. Once it is entered, an **Act now** alert opens when the air has been at or below it for two minutes: *Frost: the air at the field has fallen to the frost limit*. It can come in the night, so agree who is woken.

Settings, and where they are made

Setting	Value	Where
Unit number	**7.** Many sensors are delivered as 1, the same as another device. Set it the maker's way, with the USB-to-RS-485 adapter and no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, set the maker's way	At the office
Its label	The number, written on the sensor	At the office

Fitting

PROJECT MANAGER

- [ ] **In the shade,** a metre or two above the crop, away from walls and from the pump house. A sensor in the sun reads the sun, not the air.
- [ ] In a greenhouse: at the height of the crop, away from the heater, the vents and the door.
- [ ] It joins the data cable in the daisy chain: A to A, B to B, no branches (sheet HW03).
- [ ] Photograph where it hangs, and write the place down.

In the platform

The farm template makes the device **Weather sensor** with its two readings and no registers.

1. **Its address** (step 4, **Where each device is on the network**). The box cannot recognise the sensor, so its line says **Enter by hand**. Under the device open **Enter the address by hand** and fill in the **IP address** (the converter's: the water module's line shows it), the **Port** (502) and the **Unit number** (7). Press **Save**.
2. **Its registers** (step 5, **Which register holds which reading**). Under each reading press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx, and the address counts from zero. Under how it becomes the reading choose **Multiply and add**: a sensor that reports tenths of a degree is multiplied by 0.1. Press **Save the register**.
3. **Map the temperature as a signed number (int16).** Mapped as an unsigned one, a frosty morning reads as a hot afternoon: -2 degrees becomes thousands.
4. **The frost limit** (step 10, **Alert limits agreed with the customer**). Ask the grower at what air temperature frost damages the crop, and in which weeks. Enter it as *Frost alert at or below*, between -30 and 15 degrees.

Write what you enter:

Reading	Register kind	Address	Kind of number	Multiply by
Air temperature			int16	
Humidity				

Acceptance test

At the office, on the desk:

- [ ] The sensor has its number (7), alone on the cable, and is labelled.
- [ ] Behind the converter, both registers are read from the box, and the numbers are the room's:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 7 --type input --address 1 --data-type int16
--scale 0.1
```

(The kind, the address, the kind of number and the scale are the maker's: these are an example.)

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Weather sensor. Against a hand-held thermometer at the sensor. No tolerance is laid down: write both numbers and judge.	Plausible for the time of day, and close to the thermometer			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A frosty morning reads as a hot afternoon, or as thousands of degrees	The temperature was mapped as an unsigned number	Map it as int16 (signed), with the scale of the sensor's table
The sensor does not answer	A wrong unit number (left as delivered), a number or an address entered wrongly, a speed that is not the cable's, A and B swapped, or no power	Check each, in that order. Test the register from the box against the maker's table
Temperature and humidity are swapped	The two registers are the other way round	Read the maker's table again
A reading is ten times too big	The scale	Under **Multiply and add**, multiply by the table's factor
The air reads far hotter than the thermometer at midday	The sensor is in the sun, or by a wall	Move it into the shade, away from walls
No frost alert came on a frosty night	The frost limit was never entered	Enter the grower's own number (step 10), and send a test message to the person who is woken

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer) and R3 (a reading looks wrong).

Not yet proven

- **No temperature and humidity sensor has been read by this software, and no model is approved.** Every make is a first time: the test above is the proof, for that sensor.
- The frost alert is tested in software. It has not yet been seen on a real frosty night.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	